A causal PT decoder: the output mechanism

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Notation. $S \in \mathbb{R}^{|\mathcal{V}|\times d}$ (word–label scores), $T^{(c)} \in \mathbb{R}^{d\times d}$ (arc scores, first index dependent, second index head), $b \in \mathbb{R}^{|\mathcal{V}|}$ (word unary), $r^{(c)} \in \mathbb{R}^{d}$ (root key). Head domain $\mathcal{D}_t = \{\mathrm{ROOT}, 1, \ldots, t-1\}$. Frozen filtering marginals $\bar{q}_j \approx p(Z_j \mid w_{1:j})$ for $j < t$; contracted arc scores $B^{(c)}_{j,a} = \sum_b \bar{q}_j(b)\, T^{(c)}_{a,b}$, with $B^{(c)}_{\mathrm{ROOT},a} = r^{(c)}_a$. Temperatures λ_Z, λ_H as in the paper.

1 The rejected design, and why it failed

The design was: take $Q_i^{(T)}$ — the converged posterior over the label of word i — multiply it by a fresh matrix $W_{\mathrm{out}} \in \mathbb{R}^{d\times|\mathcal{V}|}$, softmax, and train against the shifted target w_{i+1}. Three problems, each fatal. (i) W_{out} corresponds to no factor of the graph; the output layer is external to the model whose inference the rest of the network performs. (ii) Q_i is the posterior over the label of an *already known* word, conditioned on w_i through its own unary; it answers "what is this known word's category", not "what comes next". (iii) The arc factor connects a label to the label of its *head*, never to the next position: no path in the graph leads from Z_i to the identity of w_{i+1}. The projection predicts from the wrong variable across a nonexistent edge.

2 What changed

The conditional model becomes a joint over words. Factorise left-to-right by the chain rule, each step conditioned on the *labels* of the prefix, and promote the word at slot t to a graph variable W_t:

$$
\begin{aligned}
p(w, z, h) \;&=\; \prod_{t=1}^{n} p\big(W_t, Z_t, \{H_t^{(c)}\} \,\big|\, z_{<t}\big), \\
p\big(W_t, Z_t, H_t \mid z_{<t}\big) \;&\propto\; \exp\Big(b_{W_t} + S_{W_t, Z_t} + \textstyle\sum_c \sum_{j\in\mathcal{D}_t} \mathbf{1}[H_t^{(c)}{=}j]\; T^{(c)}_{Z_t, z_j}\Big).
\end{aligned}
\tag{1}
$$

This *does* modify the graph: per slot, exactly one new node (W_t) and one new, optional, unary factor b on it. Nothing else is added. S and $T^{(c)}$ are the original matrices. The observed-word unary $\exp(S_{w_t, Z_t})$ was always the pair factor $\exp(S_{W_t, Z_t})$ with W_t clamped to the data; making the word a variable does not add that factor — it unclamps it, so S now has both indices free instead of one fixed by the observation. The prefix enters only as constants: each log-potential is linear in the one-hot encoding of z_j, so the frozen marginals contract exactly into $B^{(c)}_{j,a}$ — the same contraction as the B-matrix of Appendix B.3.1 of the paper.

3 The slot factor graph

Because the slot is a star centred at Z_t, the only route from the prefix to the vocabulary is arc factors $\to Z_t \to$ word–label factor $\to W_t$. Everything in §4 is message passing along this star.

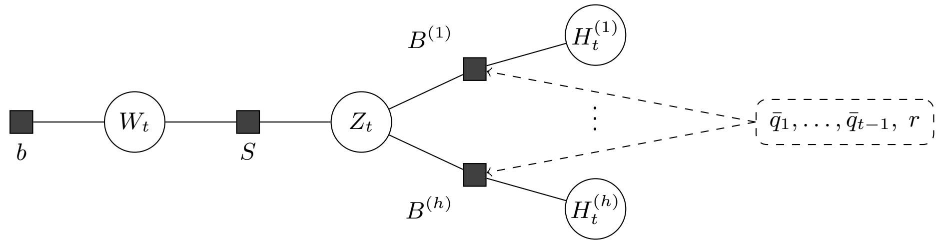

Figure 1: One slot. Circles: variables; squares: factors. The frozen prefix (dashed) enters the arc factors as constants, already contracted into the scores $B^{(c)}_{j,a}$. The graph is a star centred at Z_t.

4 The two modes

One energy per slot, two conditioning patterns. The MFVI updates of (1) are

$$Q_c(j) \;\propto\; \exp\Big(\tfrac{1}{\lambda_H} \textstyle\sum_a Q_Z(a)\, B^{(c)}_{j,a}\Big), \qquad j \in \mathcal{D}_t, \tag{2}$$

$$Q_Z(a) \;\propto\; \exp\Big(\tfrac{1}{\lambda_Z} \Big[m_W(a) + \textstyle\sum_c \sum_{j\in\mathcal{D}_t} Q_c(j)\, B^{(c)}_{j,a}\Big]\Big), \tag{3}$$

$$Q_W(w) \;\propto\; \exp\Big(b_w + \textstyle\sum_a Q_Z(a)\, S_{w,a}\Big). \tag{4}$$

Observed mode: $Q_W \equiv \delta_{w_t}$, hence $m_W(a) = S_{w_t,a}$; (4) is idle, and (2)–(3) are literally the update of the conditional chain $p(z, h \mid w)$ — the step already in place for encoding the prefix. **Predictive mode:** Q_W is free, and $m_W(a) = \sum_w Q_W(w)\, S_{w,a}$. The energy's derivative in Q_c contains no word term, so the attention update (2) is the same functional in both modes; the *only* difference anywhere is the word message, $S_{w_t,a}$ versus $\sum_w Q_W(w) S_{w,a}$. Schedule: initialise each variable from its unary-type factors (the paper's Eq. 7 rule), $Q_W^{(0)} \propto e^b$ and $Q_Z^{(0)} \propto \exp(m_W/\lambda_Z)$; run τ rounds of (2)–(3); finish with one application of (4) as the readout, $\hat{p}(W_t{=}w \mid w_{<t}) := Q_W(w)$. Take $\tau \geq 2$ so the attention query is context-dependent; at $\tau = 1$ it is the fixed probe $\sigma(\tilde{s}/\lambda_Z)$.

5 Worked example

$\mathcal{V} = \{\textit{the}, \textit{cat}, \textit{sat}, \textit{mat}\}$, $d = 2$ with labels $\{\mathrm{N}, \mathrm{V}\}$ (informally noun-like/verb-like), $h = 1$ channel, $\lambda_Z = \lambda_H = 1$, $\tau = 1$ in observed mode and $\tau = 2$ in predictive mode. Parameters (invented):

$$S = \begin{pmatrix} 1.5 & -1.5 \\ 2 & -2 \\ -2 & 2 \\ 2 & -2 \end{pmatrix} \text{ (rows } \textit{the}, \textit{cat}, \textit{sat}, \textit{mat}\text{)}, \qquad b = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 1.5 \end{pmatrix}.$$

So a noun-like dependent scores 2 under a verb-like head, a verb-like dependent scores 1 under a noun-like head, and verb-like labels are attracted to ROOT. Sentence: *the cat sat* □.

Observed pass, $t = 1, 2, 3$**.** Each slot: clamp W_t, initialise $Q_Z \propto \exp(S_{w_t,\cdot})$, one round of (2)–(3), freeze $\bar{q}_t$, cache $B_{t,\cdot} = T\bar{q}_t$.

t	w_t	$Q_Z^{(0)}$	Q_c over $\mathcal{D}_t = (\mathrm{ROOT}, 1, \ldots, t{-}1)$	$\bar{q}_t = (\mathrm{N}, \mathrm{V})$	cached $B_{t,\cdot}$
1	*the*	(.953, .047)	(1)	(.818, .182)	(.365, .818)
2	*cat*	(.982, .018)	(.414, .586)	(.957, .043)	(.085, .957)
3	*sat*	(.018, .982)	(.476, .245, .280)	(.006, .994)	(1.987, .006)

E.g. $B_{3,\mathrm{N}} = \bar{q}_3(\mathrm{N})\, T_{\mathrm{N},\mathrm{N}} + \bar{q}_3(\mathrm{V})\, T_{\mathrm{N},\mathrm{V}} = .006 \cdot 0 + .994 \cdot 2 = 1.987$: attending to the verb slot boosts the N-logit of the attender. The verb itself attends mostly to ROOT.

Slot 4, predictive. $Q_W^{(0)} = \sigma(b) = (.475, .175, .175, .175)$, so the word message is the prior-weighted mean row of S: $\tilde{s} = \sum_w Q_W^{(0)}(w) S_{w,\cdot} = (1.063, -1.063)$, giving $Q_Z^{(0)} = \sigma(\tilde{s}) = (.893, .107)$. Attention logits are $F(j) = \sum_a Q_Z(a) B_{j,a}$; e.g. round 1, $F(3) = .893 \cdot 1.987 + .107 \cdot .006 = 1.776$.

round	F over (ROOT, 1, 2, 3)	Q_c	Q_Z after (3)
1	(.160, .413, .178, 1.776)	(.120, .154, .122, .604)	(.951, .049)
2	(.074, .387, .128, 1.890)	(.104, .143, .110, .642)	(.956, .044)

Readout (4): logits $b_w + \sum_a Q_Z(a) S_{w,a}$, e.g. for *the*: $1 + .956 \cdot 1.5 + .044 \cdot (-1.5) = 2.368$.

	the	*cat*	*sat*	*mat*
logit	2.368	1.824	−1.824	1.824
$\hat{p}(W_4 \mid w_{<4})$	.460	.267	.007	.267

The context, entering only through Q_Z, puts .993 of the mass on the noun-like cluster and kills the verb; within the cluster the prior b favours *the*. Suppose *mat* is sampled.

Prior → posterior. Clamp $W_4 = mat$ and run the observed step of the *same* slot: $m_W = (2, -2)$, one round gives $Q_c = (.097, .137, .104, .663)$ and $\bar{q}_4 = (.993, .007)$, versus the predictive $Q_Z = (.956, .044)$: the word message swapped from marginalisation ($\tilde{s}$) to evidence ($S_{mat,\cdot}$) sharpens the belief (entropy .180 → .040 nats) and sharpens the attention onto the verb (.642 → .663). $\bar{q}_4$ is frozen and appended to the cache; generation continues.

6 What falls out for free

- The LM softmax is the MFVI update of W_t, Eq. (4) — an inference step, not an attached head.
- Weight tying is forced: input use (clamped $S_{w_t,\cdot}$) and output use (4) are one factor; two separate factors on the pair (W_t, Z_t) would multiply into a single matrix anyway, so untied weights are unrepresentable.
- The `[MASK]`-style query vector is the prior word message $\tilde{s} = \sum_w Q_W^{(0)}(w) S_{w,\cdot}$ with $Q_W^{(0)} \propto e^b$: the prior-weighted mean embedding, derived rather than invented.
- The KV cache is the frozen $\bar{q}_j$ (equivalently $\bar{q}_j V^{(c)}$ after $T^{(c)} = U^{(c)} V^{(c)\top}$): the cache is the filtering marginal itself.
- Two-stream attention is the clamped/free distinction: content stream = observed mode, query stream = predictive mode, on the same slot.

Each of these is standard engineering that the model derives rather than posits.

7 Costs

Stated up front. (i) Per position, two inner problems (predictive query, then observed encoding) over one shared cache: roughly 2× the attention of an equal-depth GPT, same asymptotics. (ii) All prediction flows through the d-dimensional label simplex: Z_t is the unique cut vertex between prefix and W_t. (iii) The mean-field readout is rank-d limited: logits $b_w + \sum_a Q_Z(a) S_{w,a}$ are affine in Q_Z, so across contexts the logit matrix has rank at most $d{+}1$ (a softmax bottleneck). No claim of better perplexity is made here. The claim is narrower: the output mechanism is derived from the graph, not stapled onto it.